

MYCA Orchestration Architecture — ITDX26

Scenario runtime, role resolution, and admissibility gating

Event	ITDX26 — U of Arizona Sierra Vista, 8–18 Sept 2026
Track	Algorithm-only, unclassified, laptop-based evaluation
Solicitation	W91RUSI-FCID260001 (SUB-007)
Submitter	Mycosoft, LLC · CAGE 9KR60 · UEI YK3ARVKJ77S9
PI / POC	Morgan Rockcoons, Founder & CEO · morgan@mycosoft.org · (619) 483-1077
Handling	UNCLASSIFIED // PROPRIETARY
Version	FormSpace overlay v1.0 — 2026-09-01

1. Purpose

MYCA is the mission-level orchestrator that turns a scenario definition into coordinated algorithm execution across the four ITDX tasks. It does not replace local device safety controllers, flight controllers, or the AVANI admissibility gate. Its scope is task decomposition, role assignment, hypothesis generation, COA proposal, and execution monitoring.

2. Scenario Definition Language

A scenario is a declarative mission package (schema: myca_scenario_v1). It defines identity, handling, objective, area of responsibility, safety zones, abstract roles with required capabilities, communications priorities, ordered phases, event-correlation rules, success criteria, and prohibited-language rules. A scenario that fails compilation cannot run in autonomous or semi-autonomous mode.

3. Capability-Based Role Resolution

MYCA assigns roles by capabilities, not device names. Compatibility indicator $\chi(i,r) = 1$ if role capabilities $C_r \subseteq C_i$ device capabilities, else 0. Role score $S(i,r) = w_p \cdot P + w_q \cdot Q + w_e \cdot E + w_l \cdot L - w_c \cdot C$, where P is preferred-class compatibility, Q is health, E is energy, L is link suitability, and C is tasking cost. The resolver selects the max-score feasible assignment under uniqueness and safety constraints. When a role cannot be resolved, the scenario follows its declared fallback (abort, degrade, simulate, or operator substitute); every substitution is visible in the run manifest.

4. The Admissibility Gate

MYCA never issues an action without passing through the deterministic AVANI gate $G(E, \phi, v, A, R) \rightarrow \{\text{PASS, REVIEW, DENY, PAUSE}\}$. The gate takes as inputs evidence sufficiency, policy, environmental stewardship, the action envelope, and the rollback plan. The action envelope is $A = (\text{identity, scope, magnitude, duration, preconditions, stop conditions, rollback, authority})$ — every field populated, every field visible in the operator UI. High-risk actions require explicit human disposition regardless of gate output.

5. Seasonal Governor

A state machine over $\{\text{Spring, Summer, Autumn, Winter, Frost}\}$ biases action envelopes toward the current operational season — for example, Frost tightens the reversibility requirement and reduces allowed magnitude for any deployed actuation. In ITDX evaluation the season is set from the

scenario and does not change mid-run.

6. Multi-Agent Protocol

Propose → critique → verify → score → converge or abstain → govern. Scoring uses a log-opinion pool with lineage-aware weights. Bounded to two revision iterations for time discipline. An abstention is a valid output; MYCA does not fabricate a COA to fill a slot.

7. MINDEX Persistence

Every observation, hypothesis, COA, disposition, and outcome is persisted in MINDEX with a versioned model ID H(arch, weights, data root, calibration, loss weights, code SHA, runtime, card). Outcome-linked entries record prediction, action, outcome, error, and lesson. This is the audit trail an Army SME can follow from a decision back to the physical measurement.

8. What MYCA Is Not

- Not a general reasoner. Its inputs are structured state and evidence graphs.
- Not a black-box autopilot. High-risk actions require human disposition.
- Not a replacement for local safety. Flight controllers and device safety controllers remain authoritative.
- Not an autonomous procurement agent. All external actions are logged, gated, and reversible where possible.

Morgan Rockcoons
Founder & CEO, Mycosoft, LLC
morgan@mycosoft.org · (619) 483-1077